

TAREK ELWASIF

MSc Student in Artificial Intelligence · Computer Vision & Geometric Deep Learning

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PROFILE

MSc student in Artificial Intelligence for Science and Technology at the University of Milano-Bicocca, with an undergraduate background in Electronics and Communications Engineering. Strong mathematical foundation and physics combined with hands-on experience in computer vision, transformer architectures, and probabilistic graphical models. Currently directing thesis work toward Geometric Deep Learning, applying mathematical structure — graphs, manifolds, and symmetry — to learning systems.

EDUCATION

M.Sc. in Artificial Intelligence for Science and Technology 2025 – Present
Università degli Studi di Milano-Bicocca, Milan, Italy | GPA: 27.7/30 (Year 1)

- Year 1 foundations: mathematics, physics, and statistics underpinning machine learning, followed by in-depth study of computer vision algorithms and deep neural network architectures.
- Thesis direction (in progress): Geometric Deep Learning.

B.Sc. in Electronics and Communications Engineering 2020 – 2025
Mansoura University, Mansoura, Egypt | Grade: 88/100 (Excellent)

TECHNICAL SKILLS

Programming: Python, C, C++, R

Deep Learning: PyTorch, Vision Transformers (ViT), CNNs, R-CNN, YOLO, U-Net, Transformers, Self-Supervised Learning

ML & Probabilistic Models: Scikit-learn (K-Means, Agglomerative Clustering, DBSCAN), Bayesian Networks (pgmpy), NLP, TF-IDF

Mathematics: Linear Algebra, Group Theory, Calculus, Statistics & Probability, Convex Optimisation

Tools & Platforms: NumPy, Pandas, NLTK, Git, Streamlit

ACADEMIC PROJECTS

VI-UNet — Vision Transformer Encoder with U-Net Decoder for Semantic Segmentation

MSc Course Project, Computer Vision — Università di Milano-Bicocca | 2025

- Designed a custom hierarchical Vision Transformer encoder (patch embedding, learnable positional encoding, 3-stage attention blocks) paired with a U-Net decoder for pixel-wise semantic segmentation on the CamVid road-scene dataset (32 classes).
- Benchmarked the from-scratch ViT-UNet (Val. Dice 0.66, Val. IoU 0.62) against a pretrained ResNet34-UNet baseline (Val. Dice 0.67, Val. IoU 0.63) over 30 epochs on Google Colab (T4/A100 GPUs).
- Deployed an interactive web demo of the trained model with Streamlit.

Amazon Customer Reviews — Clustering & Bayesian Network for Rating Prediction

MSc Course Project, Unsupervised Learning — Università di Milano-Bicocca | 2025

- Built an end-to-end NLP pipeline (POS-filtered lemmatisation, TF-IDF vectorisation, Truncated SVD) on 21,000+ Amazon reviews to surface latent complaint themes.
- Compared K-Means, Agglomerative Clustering (Ward linkage), and DBSCAN; hierarchical clustering recovered four interpretable complaint categories (navigation, service quality, refunds, returns).

- Learned a Bayesian Network structure (Hill-Climbing search with BIC scoring, pgmpy) over nine engineered features to model the causal dependencies behind customer ratings, achieving an MSE of 0.0908 versus a 0.222 uninformed baseline.

Hybrid CNN–Transformer Architecture for Food Image Classification

MSc Course Project, Deep Learning — Università di Milano-Bicocca | 2025

- Designed a hybrid architecture combining a ResNet-style convolutional backbone with a Vision Transformer attention stage for fine-grained classification across 251 food categories (iFood 2019), trained entirely from scratch under limited compute.
- Raised validation accuracy from ~25% (pure CNN baseline) to ~51% by combining local convolutional features with global self-attention.
- Implemented a self-supervised rotation-prediction pretext task to pretrain the encoder without labels, reaching comparable downstream accuracy in fewer epochs than full supervision.

Breast Ultrasound Tumor Detection — Malignant vs. Benign Classification (YOLOv8)

Healthcare AI Project — SI HealthCare Program

- Built a full data preparation pipeline for the BUSI (Breast Ultrasound Images) dataset, processing benign and malignant ultrasound images with their segmentation masks; extracted bounding box labels in YOLO format to support object detection training.
- Trained a YOLOv8s model (50 epochs, AdamW optimizer, 640×640 resolution) to detect and localise tumors in ultrasound scans, distinguishing malignant from benign cases.
- Implemented IoU-based evaluation utilities and bounding box conversion helpers; handled multi-mask images where a single scan contained several annotated tumour regions.
- Designed a lightweight MLP meta-model as a post-processing stage on top of the fine-tuned YOLOv8: given each predicted box and its confidence score, the network regresses coordinate deltas (Δx , Δy , Δw , Δh) to shift the box toward the ground-truth and outputs a refined IoU estimate to improve localisation precision at inference time.

AI-Driven EMG-Based Prosthetic Hand Control and Simulation System

BSc Graduation Project — Mansoura University | 2025

- Built an end-to-end system that predicts finger-joint angles from EMG (muscle signal) data via a trained ML model, streaming real-time predictions to a Blender-based 3D hand simulation over socket communication.
- Implemented an online fine-tuning feedback loop for continuous model adaptation.

Tools: Python, Blender, Socket Programming, Machine Learning

Sentiment Analysis Using Transformer Architecture (From Scratch)

AI Foundations Program — Sprints, Egypt

- Implemented the complete Transformer architecture from first principles — including custom self-attention and multi-head attention layers — for NLP-based sentiment classification.

CERTIFICATIONS & COURSES

- AI Foundations — Sprints, Egypt
- NLP with Deep Learning (CS224N) — Stanford University Online Course
- MIT 8.01/8.02/8.03 — Classical Mechanics, E&M, and Vibrations & Waves (Walter Lewin, MIT OpenCourseWare)
- The Theoretical Minimum: Quantum Mechanics & Special Relativity — Stanford University (Leonard Susskind)

LANGUAGES

Arabic	Native
English	Proficiency (C1)
Italian	Elementary (A2)